

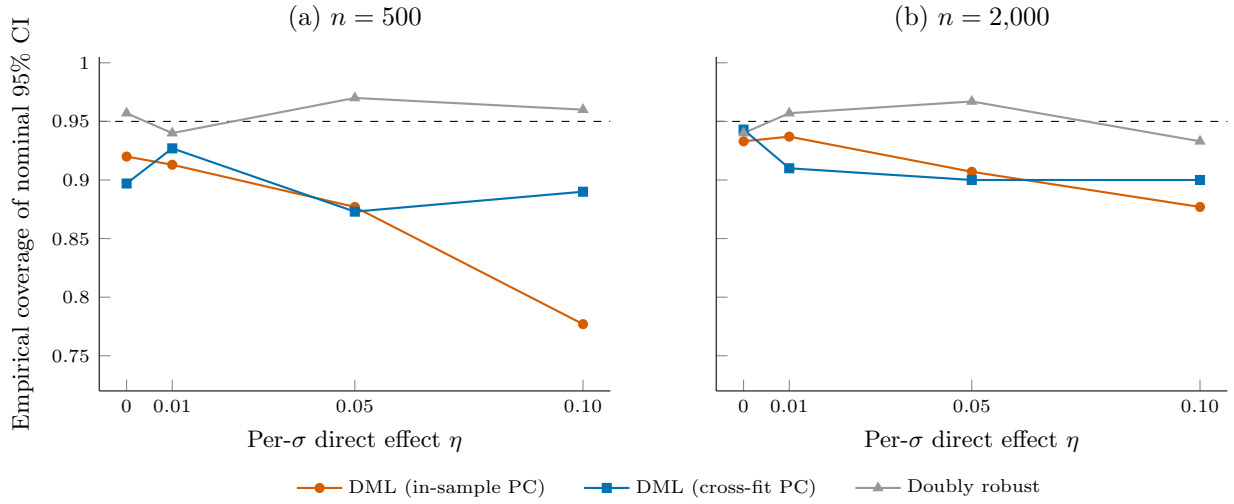


Figure 1: Finite-sample coverage of the nominal 95% influence-function interval for the embedding-valued treatment, across effect size η at $n = 500$ (a) and $n = 2,000$ (b), with 300 Monte Carlo replications per cell. The horizontal dashed line marks the nominal level. The in-sample principal-component DML interval, which treats the estimated leading component as a generated regressor, under-covers as the true effect grows, falling to 0.78 at $n = 500$, because the orthogonal-score variance does not capture the sampling variability of the estimated treatment direction. Forming the component out of fold (cross-fit PC) recovers most of the lost coverage, and the doubly-robust estimator, which does not estimate a continuous direction, is calibrated throughout but has no power against the graded effect.

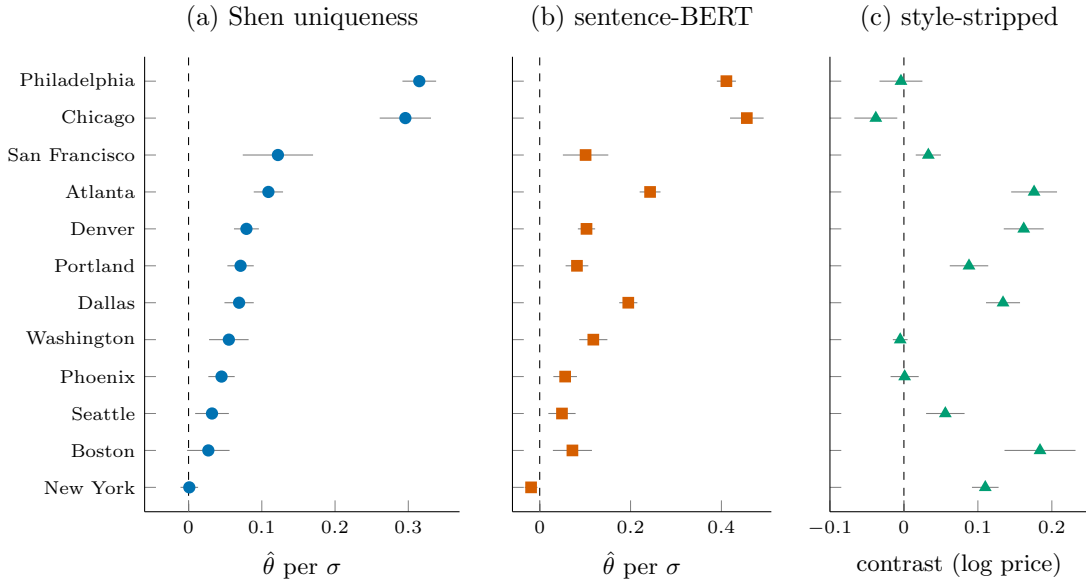


Figure 2: Per-market DML estimates across the twelve metropolitan areas for the three text channels, ordered by the Shen-Ross uniqueness effect. Filled dots are point estimates and gray whiskers are 95% intervals (influence-function for the Shen and sentence-BERT channels; coefficient-uncertainty-folded cluster bootstrap for the counterfactual contrast). The dashed vertical line marks the null. The sentence-BERT axis is oriented so its pooled effect is non-negative. Numerical values appear in Tables ??, ??, and ??.

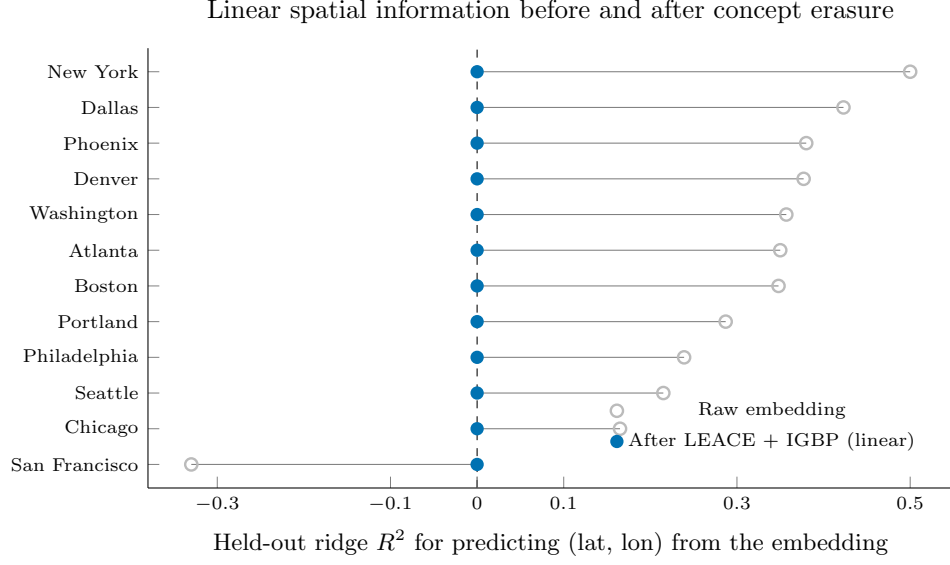


Figure 3: Held-out ridge R^2 for recovering the continuous latitude-longitude coordinate from the sentence-BERT embedding, before (open) and after (filled) the composed linear concept erasure, in each of the twelve markets ordered by raw R^2 . The raw embedding carries substantial linear spatial signal, recovering the coordinate at R^2 up to 0.50 in New York; the negative San Francisco value reflects out-of-sample probe performance below the coordinate-mean baseline rather than absent signal. Linear erasure drives the recoverable signal to within Monte Carlo noise of zero in every market, the exact-guardedness property of LEACE. Residual *nonlinear* leakage, which a linear probe cannot see, is quantified separately in Section ??.

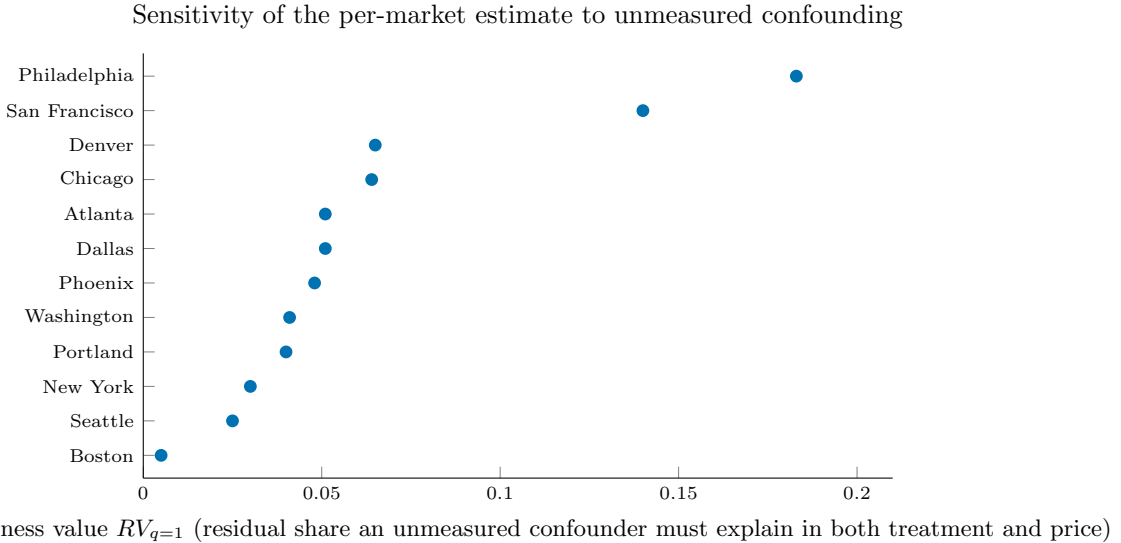


Figure 4: Robustness value for each market's sentence-BERT effect, the share of residual variance an unmeasured confounder would need to explain in both the treatment and log price to drive the estimate to zero, following ?. Larger values are more robust. The estimate is most robust in Philadelphia (0.18) and San Francisco (0.14) and most fragile in the markets whose point estimates are already near zero. The modest robustness values in several markets are the reason we read the per-market estimates as associational and sensitivity-bounded rather than cleanly causal.

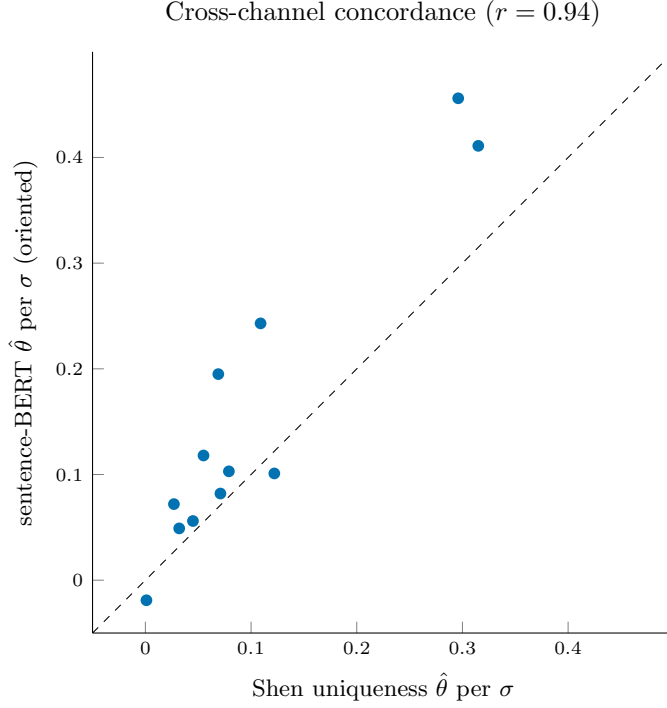


Figure 5: Per-market estimates of the two embedding channels against each other, with the 45° identity line. The two channels co-move almost perfectly across markets (Pearson $r = 0.94$), so they recover one underlying signal under two encodings rather than two independent measurements; the systematic departure from the identity line reflects that the sentence-BERT axis is roughly twice the Shen channel in magnitude, not a difference in which markets carry the signal.

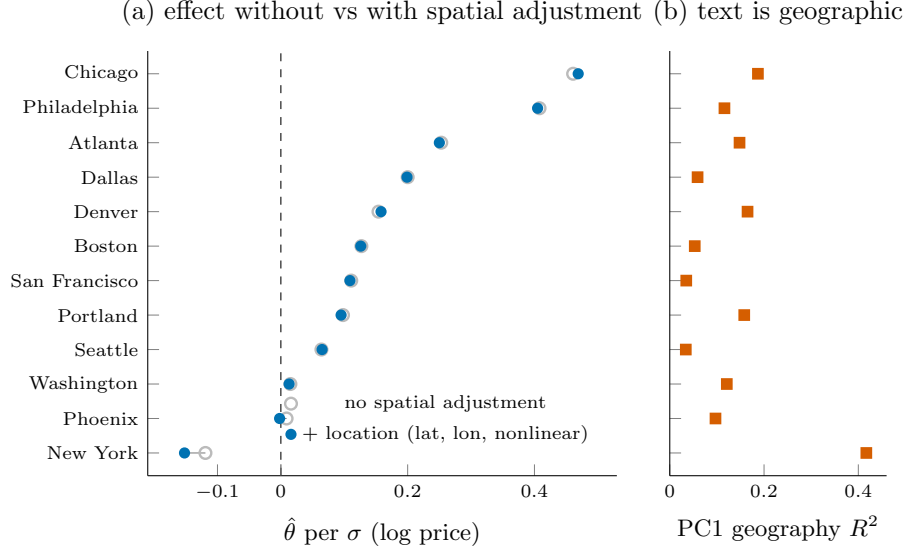


Figure 6: Is the per-market text-price effect spatially confounded? Holding the leading text principal component fixed as the treatment, panel (a) plots its DML effect on log price controlling for the structured confounders without (open) and with (filled) a location adjustment that adds latitude, longitude, and their quadratic and interaction terms to the confounder set. The two estimates nearly coincide in eleven of twelve markets, with a confounded share below five percent of the effect; New York is the exception, where the location adjustment moves the estimate by about a quarter of its magnitude without changing its sign. Panel (b) reports the share of the treatment’s own variance that the location basis explains, which reaches 0.42 in New York and exceeds 0.10 in six markets, so the robustness in panel (a) is not an artifact of a treatment that carries no geography: the text direction does encode location, yet that location component does not drive its price effect.